

POLICY REPORT

Hurricane Maria

Puerto Rico, 2017

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Glossary

AAL	Annual Average Loss
CAT	Catastrophe Risk Model
DSGE	Dynamic Stochastic General Equilibrium
EMDE	Emerging Market and Developing Economy
EP	Exceedance Probability
FEMA	Federal Emergency Management Agency
GAO	Government Accountability Office
GDP	Gross Domestic Product
IA	Individual Assistance
LCOE	Levelized Cost of Energy
NFIP	National Flood Insurance Policy
NFRIP	National Flood Risk Information Project
PDA	Preliminary Damage Assessment
PREPA	Puerto Rico Electric and Power Authority
PRERA	Puerto Rico Emergency Relief Administration
PROMESA	Puerto Rico Oversight, Management, and Economic Stability Act
SBA	Small Business Administration
TFP	Total Factor Productivity

Part 1: Catastrophe Overview

Introduction

Hurricane Maria, which struck Puerto Rico on September 20, 2017, was a catastrophic disaster that revealed deep systemic vulnerabilities in the island's economic and policy landscape. Arising at the end of an 11-year economic recession, the hurricane amplified Puerto Rico's long standing financial instability, marked by a \$72 billion USD national debt, a 10.8% unemployment rate, and a poverty rate of 45% (FEMA, 2018). These challenges were rooted in the island's economic history, including the phasing out of federal tax incentives, reliance on unsustainable borrowing, and limited political autonomy as a U.S. unincorporated territory (Meléndez, 2018). Such structural weaknesses hindered effective responses to both chronic and acute crises. This paper examines the intersection of Puerto Rico's historical economic trajectory and policy failures that exacerbated Hurricane Maria's impacts, particularly its catastrophic physical damage and deepening of social and economic inequalities. It further critiques the inadequacies of federal and local disaster responses and explores policy reforms informed by historical and international case studies to address systemic vulnerabilities, improve disaster preparedness, and promote sustainable development. Recognizing that averages can obscure concentrated inequalities, this analysis underscores the urgency of equitable and context-specific policy interventions.

Context

Puerto Rico's economic stagnation dates back to the mid-2000s, following the phased elimination of Section 936 of the federal tax code, which had incentivized U.S. corporations to operate on the island through tax exemptions (Meléndez, 2018). This policy shift triggered a severe economic contraction, leading to a prolonged recession by 2006. To counter stagnation, the government increasingly relied on public borrowing, with Governor Pedro Roselló's infrastructure initiatives in 2007 adding \$24 billion USD to the debt without fostering significant growth (Gluzmann, Guzman, and Stiglitz, 2018). Between 2006 and 2017, Puerto Rico's GDP fell by 10%, employment declined by 17%, and the population shrank by 11%, reflecting the cumulative impact of structural economic weaknesses (US GAO, 2024).

The island's fiscal challenges were compounded by its fragile political status as a U.S. unincorporated territory. While subject to federal laws, Puerto Rico lacks voting representation in Congress, curbing its fiscal autonomy and capacity to advocate for equitable federal support (McSorley, 2024). In 2008, a fiscal emergency declaration by Governor Luis Fortuño introduced austerity measures that deepened economic strain (Meléndez, 2018). By 2016, the financial crisis prompted the U.S. Congress to pass PROMESA, establishing a federal oversight board to manage the budget and restructure debt (Meléndez, 2018). However, persistent debt obligations, which reached 68% of GDP by 2017, undermined the legislation's goals, leaving the economy stagnant with a diminishing GDP per capita of \$29,800 USD (Figure 1; Trading Economics, 2024).

Puerto Rico's reliance on imports for 85% of consumption and its limited trade diversity further highlighted systemic vulnerabilities (Kenner, 2023). The island's economy, reliant on pharmaceutical exports as a major revenue source, was ill-equipped to recover from external shocks (Ferré, 2017). By

2017, Hurricane Maria compounded these vulnerabilities, adding an estimated \$43–50 billion USD in recovery costs to the existing debt (Gluzmann, Guzman, and Stiglitz, 2018). Disruptions to supply chains and fragile infrastructure magnified the island's economic distress. Puerto Rico's GDP growth rate and per capita income dropped significantly in the aftermath, underscoring their economic fragility (WB, 2024). These challenges illuminated the systemic risks of an unsustainable economic model and inadequate fiscal safeguards, necessitating structural reforms to enhance resilience and sustainable development.

Consequences

Hurricane Maria, one of the most devastating storms in U.S. history, struck Puerto Rico as a Category 5 hurricane, bringing sustained winds of 155 mph and over 38 inches of rainfall (Pasch, 2023). The storm caused catastrophic flooding, mudslides, and over 40,000 landslides, decimating critical infrastructure (Pasch, 2023). Over 80% of utility poles and 95% of transmission lines were damaged, resulting in the most extensive power outage in U.S. history, leaving 3.4 million residents without electricity (FEMA, 2024). Nearly 357,500 homes were damaged, and the island's communications network was almost entirely destroyed, exacerbating the disaster's impact (Kenner, 2023). Prolonged infrastructure failure contributed to an 18% vacancy rate in Puerto Rico's housing stock, underscoring the storm's severe physical toll (Kenner, 2023).

Economically, Hurricane Maria was the third costliest hurricane in U.S. history, cutting Puerto Rico's economic output by 20% and projecting a 21% reduction in income over the next 15 years—equivalent to \$180 billion in cumulative losses (Figure 2; Ferré, 2017). The agricultural sector, already fragile due to years of economic stagnation, suffered an 80% loss in crop value, equating to \$780 million USD in agricultural yields, and a 37.5% decrease in operational farms (Kenner, 2023). Puerto Rico's balance of trade also plummeted, shifting from a \$1.47 billion USD surplus in August 2017 to a \$923.8 million USD deficit by May 2018, reflecting the economy's heavy dependence on imports and limited trade diversification (Figure 3; Trading Economics, 2024). These economic shocks compounded pre-existing vulnerabilities in a territory reliant on trade and federal aid. GDP per capita, already constrained before the storm, further declined in 2017 and 2018, highlighting the hurricane's far-reaching economic impacts (WB, 2024).

Socially, the storm's aftermath triggered a widespread health and well-being crisis. Extended power outages and disrupted access to essential services exacerbated inequalities, pushing over 50% of residents previously above the poverty line into poverty (Lu, 2018). Nearly 3,000 deaths were directly or indirectly linked to the hurricane, emphasizing the human cost of systemic vulnerabilities (FEMA, 2018). Moreover, Hurricane Maria catalyzed a mass migration, reducing Puerto Rico's population by 14% within a year (Meléndez, 2018). This exodus predominantly saw overrepresentation of young workers and child-rearing families, leading to a weakened labour market and further economic destabilization (Meléndez, 2018). Population decline, which had been ongoing since 2007, reached a critical low after the storm, and the loss of farm workers significantly undermined the agricultural sector, a key component of Puerto Rico's economy (Ferré, 2017).

Part 2: Policy Failures

I. NFIP

The devastation of Hurricane Maria exposed critical policy failures within the NFIP. Established to provide affordable flood insurance and manage risk, the NFIP failed to adequately serve Puerto Rico, leaving the majority of its residents unprotected against the financial impacts of the disaster (Kousky and Lingle, 2018). At the time of the hurricane, only 4% of Puerto Rican households were enrolled in the NFIP, a strikingly low figure compared to the national average (Figure 4; Kousky and Lingle, 2018). This lack of coverage underscores systemic issues with the program's design and implementation, specifically in areas like Puerto Rico that are highly vulnerable to natural disasters.

One major reason for low enrollment is the NFIP's limited applicability. Federal requirements for flood insurance are restricted to households in designated flood zones with a 1% annual chance of flooding and holding federally-backed mortgages (FEMA, 2018). However, less than 30% of Puerto Rican households carry such mortgages, and nearly half of the island's residents live in informal or technically "illegal" housing, in fact only 41% of owner occupied homes were mortgaged when Maria hit (Atreya et al., 2015). These factors effectively exempt large portions of the population from mandatory flood insurance, leaving them exposed to catastrophic losses.

Further compounding the issue are outdated flood hazard maps produced by FEMA, which determine NFIP's coverage requirements. These maps often fail to incorporate critical climate data, such as accurate rainfall projections, leading to significant underestimations of flood risk (Kousky and Lingle, 2018). Independent risk models indicate that over three times as many homes in Puerto Rico face high flood risks compared to FEMA's assessments (Kousky and Lingle, 2018). As a result, many households that should qualify for NFIP policies are excluded, reducing the program's effectiveness in disaster-prone areas.

The shortcomings of the NFIP have reignited debates on the role of public versus private insurance in managing disaster risks. Public insurance, like the NFIP, offers advantages in geographic diversification and solvency potential, spreading risk across unaffected regions and drawing on taxpayer-backed funds for payouts (Wagner and Marcoux, 2023). This model is particularly suited to large-scale events like hurricanes, which can overwhelm the localized resources of private insurers (Figure 5; Kousky and Lingle, 2018). In contrast, private insurers often operate within narrow, high-risk markets, such as Puerto Rico (Wagner and Marcoux, 2023). When disasters strike, these insurers face a concentrated surge in claims, increasing their vulnerability to insolvency and limiting their ability to provide adequate coverage (Atreya et al., 2015).

However, the NFIP's challenges highlight the limitations of public programs. Despite a \$16 billion USD debt forgiveness in 2017, the NFIP remains over \$20 billion USD in debt, frequently relying on Treasury loans to fund payouts (Kousky and Lingle, 2018). Underinsured properties in high-risk zones exacerbate this financial strain, as collected premiums are insufficient to cover escalating losses (Wagner and Marcoux, 2023). The program also faces issues of moral hazard, where the availability of insurance reduces incentives for proactive risk mitigation. In Puerto Rico, reliance on post-disaster payouts rather

than investments in flood defenses or resilient infrastructure has increased the financial burden on the NFIP and left communities vulnerable to recurring crises.

II. The Stafford Act

The Stafford Act, officially the Robert T. Stafford Disaster Relief and Emergency Assistance Act, is a U.S. federal law aimed at providing systematic disaster relief to state and local governments. However, the response to Hurricane Maria in Puerto Rico revealed profound policy failures and systemic inequities in its implementation (*Rebuild Puerto Rico*, 2021). While the Act authorizes federal disaster aid through emergency or major disaster declarations, its application in Puerto Rico was hampered by mismanagement, inefficiency, and structural challenges that disproportionately affected the territory compared to mainland U.S. states.

Despite the Stafford Act entitling Puerto Rico to federal disaster aid, recovery efforts were significantly slower and less comprehensive than those for Hurricanes Harvey and Irma, which struck Texas and Florida the same year (Willison et al., 2019). Within the first nine days post-landfall, survivors of Harvey and Irma had received nearly \$100 million USD each in FEMA aid, whereas Puerto Rico had only received \$6 million USD; by two months, federal assistance for Harvey and Irma approached \$1 billion USD, but Puerto Rico's funding did not reach that milestone until four months post-landfall (Figure 6; Willison et al., 2019). This disparity was not reflective of the hurricane's severity—Maria caused \$90 billion USD in damages—highlighting inequitable federal resource allocation.

FEMA's mismanagement under the Stafford Act further exacerbated Puerto Rico's recovery challenges. The agency lost visibility of approximately 38% of commodity shipments to the island, mismanaging an estimated \$257 million USD (Kousky et al., 2023). Shipments took an average of 69 days to reach their destinations, and FEMA failed to implement basic tracking mechanisms such as GPS transponders or documented proof of delivery (Kousky et al., 2023). This lack of planning and adherence to established procedures undermined the timely delivery of life-sustaining resources, leaving Puerto Ricans without essential supplies.

The Stafford Act's IA program also failed Puerto Ricans. Approximately 60% of applications for FEMA aid were denied due to residents' inability to provide formal homeownership documentation, a common issue given that nearly half of Puerto Ricans live in informal housing (Willison et al., 2019). Although FEMA's guidelines allow for document exceptions, inconsistencies in their application created systemic barriers to assistance. Of the 1.1 million applications submitted, only 441,000 were approved by March 2018, six months after Maria's landfall (Meléndez, 2018).

These challenges were compounded by centralized decision-making and poor communication between FEMA, local municipalities, and the federal government. The allocation of resources was uneven, with federal staffing levels in Puerto Rico lagging behind those in Texas and Florida (Willison et al., 2019). For instance, within nine days of landfall, 30,000 federal employees were deployed in Texas, compared to just 10,000 in Puerto Rico. At its peak, Puerto Rico had 19,000 federal personnel, well below Texas's 31,000 (Willison et al., 2019).

The inefficiencies of the Stafford Act response in Puerto Rico were further highlighted by its support for the NFIP. Of the \$16 billion USD allocated to the NFIP after Hurricane Maria, only \$121,000 USD reached policyholders in Puerto Rico, as most residents were ineligible due to low enrollment rates and stringent policy criteria (Meléndez, 2018).

III. PREPA

The hurricane also revealed the deep-seated failures of PREPA, rooted in its hierarchical, centralized energy system, political corruption, and decades of financial instability. The disaster triggered the longest blackout in U.S. history, leaving some residents without power for 328 days (Figure 7; Lu and Alcantara, 2018). The collapse of both energy and communication infrastructure left Puerto Ricans vulnerable and isolated, with devastating social and economic repercussions.

PREPA's reliance on fossil-fuel-based, centralized generation exemplifies an outdated energy model ill-suited to Puerto Rico's abundant solar resources (O'Neill-Carillo and Rivera-Quinones, 2018). Technical weaknesses compounded these systemic issues, with the bulk of generating capacity concentrated in the island's south, while the highest demand occurred in the north (O'Neill-Carillo and Rivera-Quinones, 2018). Transmission lines, critical to the economy and daily life, lacked the resilience to endure major disruptions, leaving the grid dependent on external support. PREPA's centralized structure and immutable attitude toward reform alienated stakeholders and reinforced inefficiencies.

Financial mismanagement has long plagued PREPA. Decades of dependence on municipal bonds to artificially maintain rate structures, coupled with an inability to adapt during economic shifts such as the 2006 repeal of Section 936, led to its 2017 bankruptcy and \$9 billion USD debt (Velázquez, 2022). Policy decisions, such as prioritizing bond payments over critical maintenance, undermined grid resilience. Moreover, PREPA's political entanglements further impeded reform, with frequent interference from stakeholders focused on short-term gains rather than sustainable energy development (Gluzmann, Guzman, and Stiglitz, 2018).

Efforts to transition Puerto Rico's energy system to renewable and sustainable sources have repeatedly failed. PREPA's lobbying against integrating disconnected private systems and its dominance of centralized planning marginalized alternative approaches (O'Neill-Carillo and Rivera-Quinones, 2018). Although reforms like Act 57-2014 aimed to promote renewable energy and reduce bureaucratic barriers, PREPA's financial instability and centralized governance prevented meaningful implementation (O'Neill-Carillo and Rivera-Quinones, 2018). Initiatives encouraging public empowerment, such as regional agreements with NGOs and academic organizations, failed due to lack of coordination and resources.

Maria exposed the urgent need to decentralize Puerto Rico's energy system. A distributed model utilizing microgrids could increase infrastructure resiliency, particularly in isolated regions (O'Neill-Carillo and Rivera-Quinones, 2018). While the island has achieved grid parity for solar energy, systemic failures have hindered its widespread adoption. Moving beyond PREPA's centralized, fossil-dependent infrastructure is not only a technical necessity but also a socio-economic opportunity, as locally developed energy systems can stimulate economic activity through small- and medium-sized

enterprises. Ultimately, PREPA's failures reflect a convergence of narrow planning visions, susceptibility to political and economic power, and resistance to reform. Addressing these issues requires an inclusive approach to energy policy that prioritizes resilience, decentralization, and community engagement.

Part 3: Recommendations

I. NFIP

A. Leveraging Stress Testing and Tailored Risk Assessments for Insurers

Integrating advanced stress testing methodologies, as outlined by Hallegatte et al. (2022), offers a robust framework for enhancing the NFIP's ability to anticipate and mitigate future flood risks. These methodologies employ localized climate models, asset-specific data, and insurance-standard vulnerability functions to evaluate probabilistic risks, delivering a more nuanced understanding of potential damages (Hallegatte et al., 2022). For Puerto Rico, the application of such models could significantly refine the assessment of physical capital damage rates and inform premium pricing based on dynamic and localized climate scenarios. The importance of this approach is underscored by the NFIP's failure to predict the scale of destruction caused by Hurricane Maria, which left the majority of Puerto Rico's population without adequate insurance coverage.

A key recommendation involves the incorporation of localized climate models that project flood frequency and intensity specific to Puerto Rico. These models should draw from country-specific climate science, supported by expanded policies mandating enhanced climate research tailored to regional vulnerabilities (Hallegatte et al., 2022). Additionally, probabilistic risk assessments using CAT models can provide critical metrics such as AALs and EP curves, aiding scenario planning and informing insurance design. For example, stress testing frameworks like those developed by the IMF could be expanded to account for climate-related risks, ensuring that both private insurers and federal programs can withstand macro-financial shocks (IAIS, 2003). These models have proven effective in other disaster-prone regions, such as the Philippines, where typhoon stress tests revealed potential GDP reductions of 5–14%, demonstrating the systemic risks often overlooked in traditional evaluations (Hallegatte et al., 2022).

The Hallegatte et al. (2022) framework, applied in EMDEs, employs four interconnected submodules—climate scenarios, disaster scenarios, DSGE macro scenarios, and stress testing modules—integrating exposure data and vulnerability functions to better predict future losses and guide policy decisions (Hallegatte et al., 2022). By calculating variables through this model-based analysis, disasters can be viewed through both decreased and misallocated capital, ultimately representing a decline in TFP, which can have dramatic and negative impacts on GDP. Tailoring these probabilistic assessments to Puerto Rico's unique conditions allows policymakers to close critical knowledge gaps, enhance flood insurance coverage, and improve resilience planning.

Adopting these methods within Puerto Rico's flood insurance system would address systemic failures, align premiums with real risks, and foster a more resilient financial framework for managing

natural disasters. Through evidence-based modelling and expanded policy frameworks, Puerto Rico could build an insurance system capable of better mitigating the profound economic and social impacts of future climate events.

B. A Comparative Analysis of Policy Structures

Flood insurance frameworks differ globally, encompassing public-private partnerships, voluntary versus mandatory coverage, and standalone versus bundled policies. The work of Atreya et al. (2015) underscores the need to address these structural variations to enhance accessibility and effectiveness. In the U.S., flood insurance employs partial risk-based pricing, accounting for hazard and exposure but not vulnerability, limiting its ability to equitably serve high-risk areas like Puerto Rico (Figure 8; Wagner and Marcoux, 2023). To avoid crowding out private insurance and to expand coverage to households without federally-backed mortgages, policy reforms must incentivize public-private collaboration.

One critical area of reform is mandatory versus voluntary insurance. Currently, U.S. regulations require flood insurance only for properties in high-risk zones with federally-backed mortgages (Atreya et al., 2015). This structure excludes a significant portion of Puerto Rican households, where federally-backed mortgages are rare, and informal housing is prevalent (Kenner, 2023). Expanding mandates to account for these contextual realities could reduce adverse selection and broaden coverage. Furthermore, Puerto Rico relies on standalone flood insurance policies, which limit participation. Bundling flood insurance with homeowners' insurance would streamline claims processes, lower administrative costs, and encourage enrollment (Atreya et al., 2015).

Public-private partnerships provide a blueprint for reducing systemic risk and ensuring affordability through governments that offer reinsurance to mitigate private sector exposure, promoting a sustainable insurance market. Puerto Rico, which has characteristics of an uninsurable market under purely private systems, could benefit from adopting a similar coordinated approach (Kousky and Lingle, 2018). Currently, FEMA's NFIP underwrites catastrophic risk while private insurers manage policies. Strengthening this model through increased collaboration, particularly in claims management and risk awareness campaigns, could enhance resilience.

Additionally, incentivizing risk mitigation measures, such as flood-proofing, through premium discounts or increased coverage limits would align policy incentives with resilience-building efforts (Dufty, Dyer, and Golnaraghi, 2020). Such reforms could transform Puerto Rico's flood insurance system, ensuring broader participation, equitable access, and greater climate adaptability.

C. Australian Flood Risk Management and Resilience Building

Australia's proactive approach to flood risk management offers valuable insights for improving Puerto Rico's flood insurance system. As highlighted by Dufty et al. (2020), Australia's strategy integrates advanced flood mapping, preventative measures, and robust data-sharing initiatives supported by significant government investment (Figure 9; Dufty, Dyer, and Golnaraghi, 2020). The NFRIP provides accessible, high-quality flood data, empowering insurers, planners, and communities to make informed decisions (Dufty, Dyer, and Golnaraghi, 2020). Complementing this, preventative policies such as house-raising schemes, land swaps, and minimum floor level requirements mitigate vulnerability.

Regulatory frameworks, such as Queensland’s State Planning Policy, enforce building codes and designate flood hazard zones, creating a strong foundation for resilience (Dufty, Dyer, and Golnaraghi, 2020).

Puerto Rico can adapt these strategies by prioritizing investments in flood mapping and incorporating advanced risk assessments into NFIP pricing models. Open data initiatives and public education campaigns could enhance community awareness and preparedness, fostering a culture of proactive risk management. Economically, Australia’s emphasis on prevention and preparedness has demonstrably reduced post-disaster recovery costs and insured losses. Implementing similar measures in Puerto Rico would not only strengthen the island’s resilience but also reduce long-term NFIP liabilities, contributing to economic stability and sustainable recovery efforts.

D. Addressing Systemic Barriers to NFIP Reform

A more effective and inclusive flood insurance framework for Puerto Rico requires targeted reforms that address systemic barriers while leveraging robust data to mitigate the need for costly post-disaster bailouts. Key challenges include regulatory delays, capacity constraints, and affordability concerns (Hallegatte et al., 2022). Streamlined approvals, such as pre-approved contracts and expedited financial processes, can accelerate post-disaster reconstruction. Investments in skill development programs are essential to bridging technical capacity gaps in construction and flood management. To ensure equitable access, means-tested subsidies should be implemented to reduce premium burdens on economically disadvantaged households (Martin et al., 2023).

These measures, combined with investments in advanced data systems for risk modelling and pricing, can enhance the efficiency of the NFIP while supporting resilience in vulnerable communities. By integrating these strategies, Puerto Rico can move toward a sustainable flood insurance system that aligns with broader socio-economic stability and reduces dependence on costly ad hoc crisis responses.

II. Stafford Act

A. Improving Transparency, Publicization, and Quantification

The Stafford Act lacks mechanisms for consistent transparency and quantification, leading to inequitable disaster responses. Disparities in emergency response underline the urgent need for FEMA to publicize disaster response timelines, staffing, and funding levels. Additionally, the lack of standardized criteria for PDAs and disaster declarations exacerbates these inequities. Disaster declarations, critical for resource allocation, are unilateral presidential decisions influenced by political considerations, rather than being grounded in objective assessments of damage and need (Figure 10; Kousky et al., 2023). FEMA’s internal assessments have also proven inconsistent, with inspectors often inexperienced, incentivized to move quickly, or biased in their evaluations, disproportionately disadvantaged low-income households (Kousky et al., 2023).

To address these issues, federal disaster policy should incorporate social vulnerability metrics into PDAs and ensure transparent and consistent damage assessments. Investments in advanced monitoring technologies, enhanced training for FEMA assessors, and centralized oversight of state-level assessments

are crucial. Furthermore, a unified federal disaster application process could streamline resource allocation and reduce the administrative burden on Puerto Ricans.

B. Expanding to Include Long-Term Policy Beyond Immediate Rebuilding

The Stafford Act's narrow focus on emergency relief perpetuates vulnerabilities by mandating reconstruction of infrastructure to its pre-disaster state, disregarding the opportunity to build resilient systems (Martin et al., 2023). In Puerto Rico, this policy perpetuated reliance on a fragile, centralized electrical grid, leaving energy costs high and infrastructure prone to future disruptions. Federal funds were prohibited from modernizing the energy infrastructure, despite the glaring need for resilience (US GAO, 2024).

Long-term disaster recovery policies must prioritize sustainable infrastructure investments. Policies should incentivize local governments to adopt stricter building codes and promote resilient infrastructure, addressing both pre-existing vulnerabilities and future risks (Martin et al., 2023). Federal assistance programs should clarify objectives, shifting from immediate repairs to comprehensive, community-focused rebuilding strategies.

Integrating equity measures into these policies is essential. Puerto Rico's lack of political representation in the Senate and limited influence in federal funding decisions underscores the need for tailored approaches (Martin et al., 2023). FEMA's 2019 initiative to collect "disaster-impacted population profiles" must evolve into a comprehensive framework that transparently informs disaster declarations and resource allocations (Kousky et al., 2023). Congressional parameters should guide cost-sharing waivers to minimize politicization and focus on community needs.

C. Improving Consistency and Addressing US Territories Through Executive Order 13985

The inconsistency in federal disaster response highlights systemic biases rooted in political considerations. Research shows states with congressional representation on FEMA oversight committees and those politically important to the sitting president receive disproportionate aid (McSorley et al., 2024). This inequity is evident in the aid-to-damage ratio after Hurricane Maria (Willison et al., 2019). The Stafford Act must be reformed to ensure equitable treatment of all US territories, particularly those with limited political representation like Puerto Rico. Executive Order 13985, which addresses racial equity and underserved communities, provides a starting point but lacks intentional language specific to unincorporated territories (McSorley et al., 2024). An expanded executive order tailored to FEMA should explicitly mandate equitable aid distribution to all US states and unincorporated territories. This would help address the structural disparities that have historically disadvantaged Puerto Rico.

Consistency in disaster policy also requires reevaluating procedural inequities. FEMA's IA program, for instance, which denied 60% of Puerto Rican applicants due to lack of ownership documentation despite existing guidance allowing exceptions, glaringly highlights the ineffectiveness of current processes (Martin et al., 2023). Streamlining and simplifying application processes through measures like pre-registration for FEMA and SBA disaster loans during tax filing could ensure fairer access to aid.

Additionally, the practice of congressional supplemental appropriations, which accounts for 75% of disaster funding, should be revisited (Kousky et al., 2023). Establishing objective criteria for funding allocations and incorporating public input could reduce political influence and improve consistency. FEMA's risk management plans, approved by the GAO but lacking consistent monitoring, should be fully implemented to address systemic inefficiencies.

These recommendations—improving transparency, expanding long-term policies, and addressing inequities through targeted executive action—are critical to reforming the Stafford Act. By addressing the failures exposed during Hurricane Maria, policymakers can promote equitable disaster response and recovery, ensuring that all US states and territories are treated fairly in times of crisis.

III. PREPA

A. Decentralizing Governance and Infrastructure

The post-Hurricane Maria context underscores the urgent need for a transformative energy policy for Puerto Rico, centred on decentralization, and community-driven governance. PREPA's entrenched dependence on centralized, fossil-based energy systems has proven unsustainable, and reform must prioritize resilience, economic stability, and public empowerment. PREPA's hierarchical structure and centralization of power generation in the south, despite economic demand in the north, have exposed critical vulnerabilities (O'Neill-Carillo and Rivera-Quinones, 2018). The legacy of centralized electrification can be traced back to the 1935 establishment of PRERA as a public power company, founded on the principle that economies of scale reduce electricity costs. However, this foundational premise also represents a critical weakness, as it entrenches PREPA in hierarchical control, centralized generation, and top-down planning (O'Neill-Carillo and Rivera-Quinones, 2018). A decentralized energy system would improve post-disaster recovery by reducing reliance on fragile transmission lines. Establishing independent regional energy entities with decision-making authority and clear regulatory oversight can ensure the equitable distribution of resources (O'Neill-Carillo and Rivera-Quinones, 2018). This aligns with the Stafford Act by creating resilient infrastructure that minimizes future federal relief costs. Stakeholder engagement, including NGOs, local governments, and community organizations, should be institutionalized to democratize energy policy development.

B. Exploring Energy Alternatives and Resiliency Options

Transitioning to renewable energy in Puerto Rico requires a pragmatic and balanced approach, acknowledging both the potential benefits and significant challenges. A complete shift from fossil fuels to renewables risks exacerbating Puerto Rico's already high electricity costs, undermining energy security, and exposing vulnerabilities in grid resilience. Policies should prioritize flexible, cost-effective energy systems tailored to the island's unique geographic and economic context.

Renewable-based microgrids offer decentralized energy solutions, particularly valuable in rural and isolated areas prone to grid disruptions. Integrating solar photovoltaics (PV) with energy storage systems can provide localized backup power, improving reliability. However, evidence from new research in Australia highlights critical challenges: renewable energy systems often involve staggeringly high costs and have difficulty maintaining grid stability (Uhlmann, 2024). Converting intermittent renewable energy

generation into a reliable power supply places significant strain on infrastructure and requires expensive investments in system balancing (Baggu et al., 2024).

In Puerto Rico, these challenges are compounded by its unique geographic constraints, including a lack of interconnection with mainland grids and limited land availability (The Breakthrough Institute, 2024). The LCOE for renewables remains significantly higher than on the U.S. mainland, driven by land costs, weather risks, and import dependencies (Figure 11; Kuhl et al., 2024). These realities underscore the need for cautious planning to avoid overly optimistic assumptions that could exacerbate economic burdens on residents already paying electricity rates twice as high as the U.S. average.

Immediate upgrades to Puerto Rico's power infrastructure are essential. Short-term priorities should include deploying storage at critical points, improving voltage regulation, and upgrading vulnerable distribution feeders to enhance resilience against extreme weather events (Kuhl et al., 2024). Grid-forming inverters and updated operational strategies are critical to support the integration of renewable energy while ensuring system stability (Kuhl et al., 2024). Rooftop solar PV and battery systems should serve as complementary backup solutions rather than primary replacements for centralized generation.

A fully renewable grid, while aspirational, may not be economically or technically viable in the near term (Figure 12). Research shows that renewable energy alone often cannot provide the frequency regulation and stability required for reliable grid operation (Figure 13; Baggu et al., 2024). Policymakers should consider diversifying decarbonization approaches, including low-carbon nuclear energy, which offers a stable and cost-effective alternative. Efficiency retrofits in fossil fuel-based systems can also reduce emissions during the transition to cleaner energy.

By integrating renewable microgrids thoughtfully, addressing resilience challenges, and diversifying energy technologies, Puerto Rico can work toward a sustainable energy future without sacrificing affordability or stability. These actions will ensure a balanced approach to decarbonization that prioritizes the needs of its residents and industries. Additionally, these measures will support the pharmaceutical industry, a cornerstone of Puerto Rico's economy that depends significantly on fossil fuels for production and manufacturing.

These recommendations require robust and transparent regulatory frameworks to mitigate political interference and ensure implementation fidelity (O'Neill-Carillo and Rivera-Quinones, 2018). By decentralizing governance, investing in renewable-based microgrids, and fostering local economic development, Puerto Rico can rebuild its energy infrastructure to be resilient, sustainable, and community-oriented. This transformation will not only address the systemic failures exposed by Hurricane Maria but also position Puerto Rico as a leader in distributed energy innovation.

Conclusion

Hurricane Maria starkly illuminated Puerto Rico's systemic vulnerabilities, highlighting the intersection of economic fragility, policy failures, and social inequities. The island's long-standing

financial instability, limited political autonomy, and centralized governance structures compounded the hurricane's devastating impacts, leaving it poorly equipped to recover. Failures in critical programs like the NFIP and the Stafford Act revealed inadequacies in addressing Puerto Rico's unique needs as a U.S. territory, while the collapse of PREPA underscored the consequences of mismanagement and outdated local energy systems. Addressing these challenges requires comprehensive reforms. Recommendations include leveraging tailored risk assessments and stress testing within the NFIP, enhancing transparency and equity in Stafford Act implementation, and decentralizing PREPA's governance while investing in renewable energy and microgrid technologies. By adopting lessons from international case studies and prioritizing resilience and equity, Puerto Rico can move toward sustainable recovery, addressing its structural challenges to build a more secure future in the face of future disasters.

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Appendix

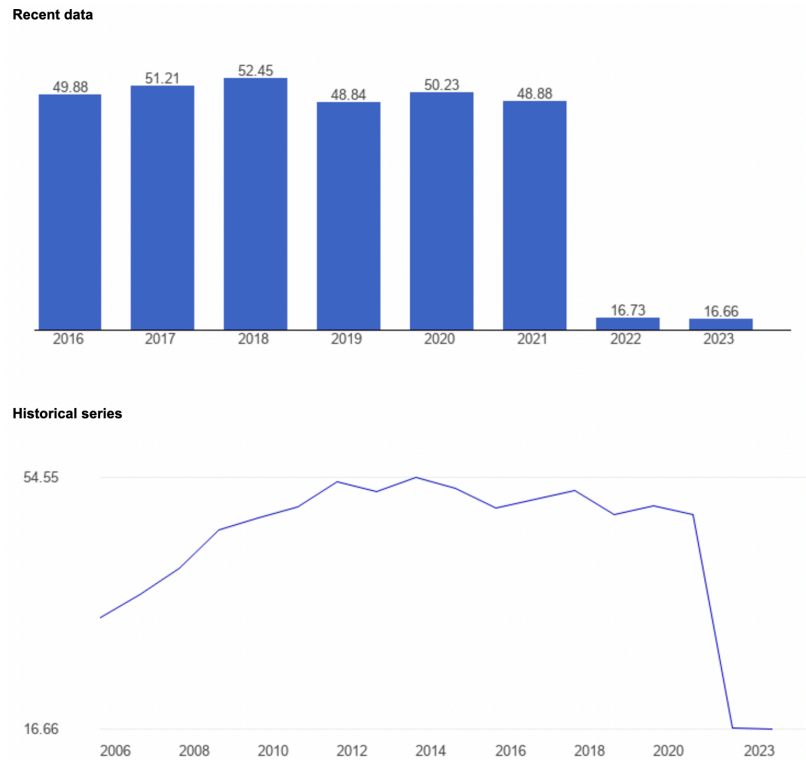
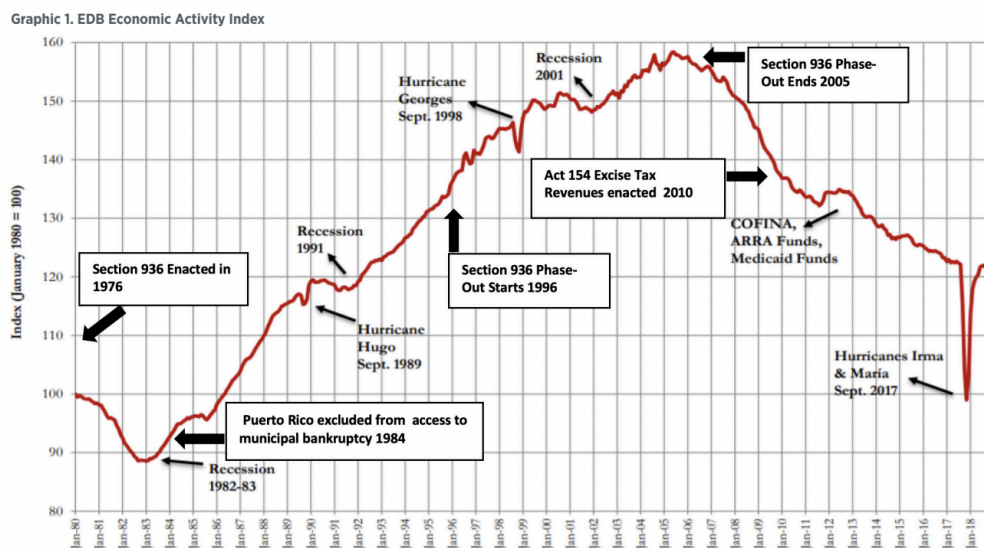


Figure 1: Government Debt as a Percentage of GDP, 2016-2023



Source: The Economic Development Bank for Puerto Rico (EDB) The Puerto Rico Economic Activity Index ("EDB-EAI") November 2018.

Figure 2: Economic Activity Index for Puerto Rico, 1980-2018

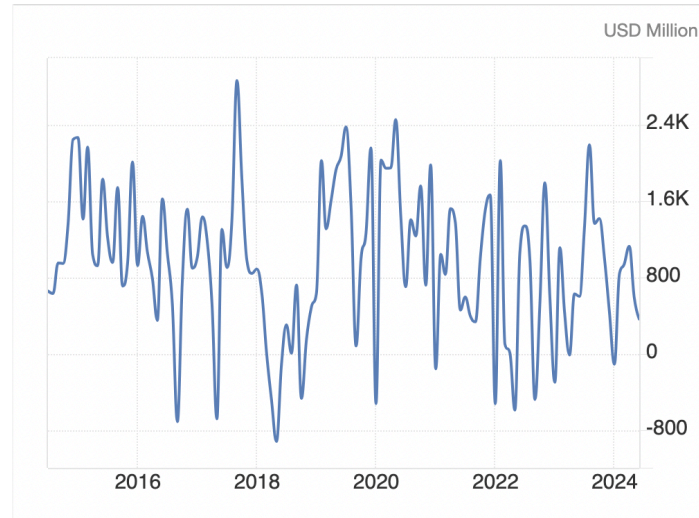
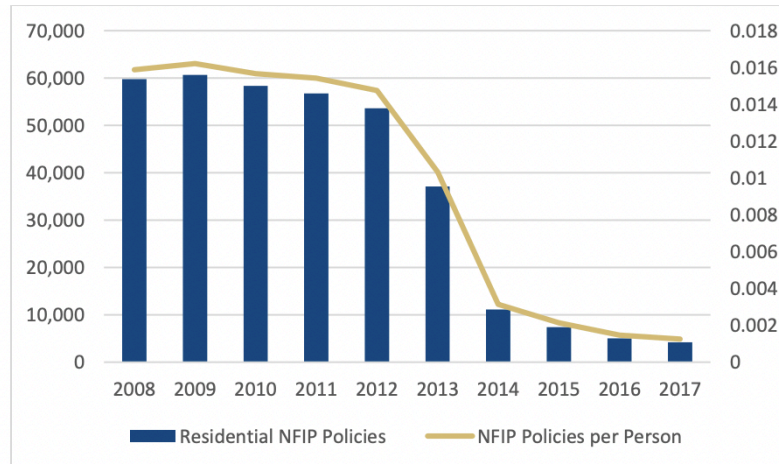


Figure 3: Balance of Trade for Puerto Rico, 2014-2024



Produced by the authors with data from FEMA and the U.S. Census.

Figure 4: NFIP Enrollment for Puerto Rico, 2008-2017

Rank	Entity	Direct Premiums Written	Market Share (%)
1	MAPFRE SA	\$14,345,000	73.80%
2	United Surety & Indemnity Co.	\$4,215,000	21.69%
3	FM Global	\$730,000	3.75%
4	Allianz Group	\$116,000	0.60%
5	Swiss Re AG	\$26,000	0.13%
6	Alleghany Corp.	\$6,000	0.03%

Data from S&P Global Market Intelligence.

Figure 5: Private Insurers in Puerto Rico, 2016

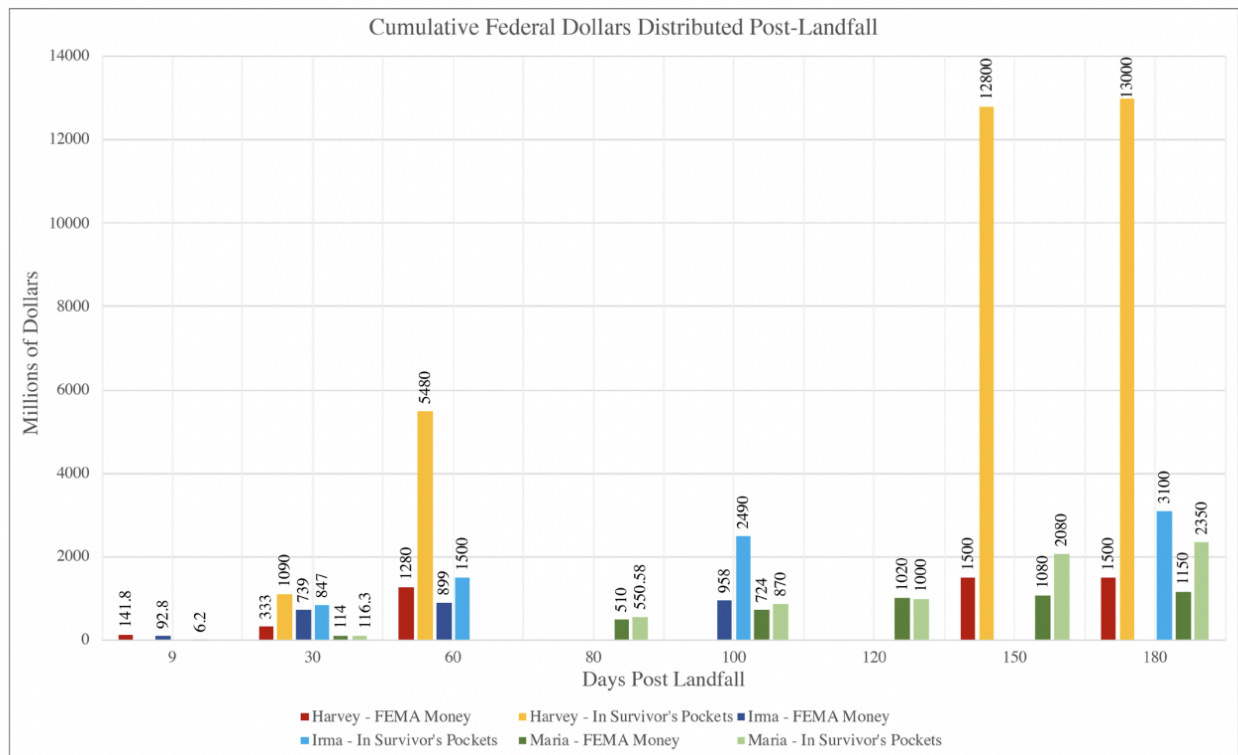


Figure 6: Cumulative Federal Dollars Distributed Post-Landfall

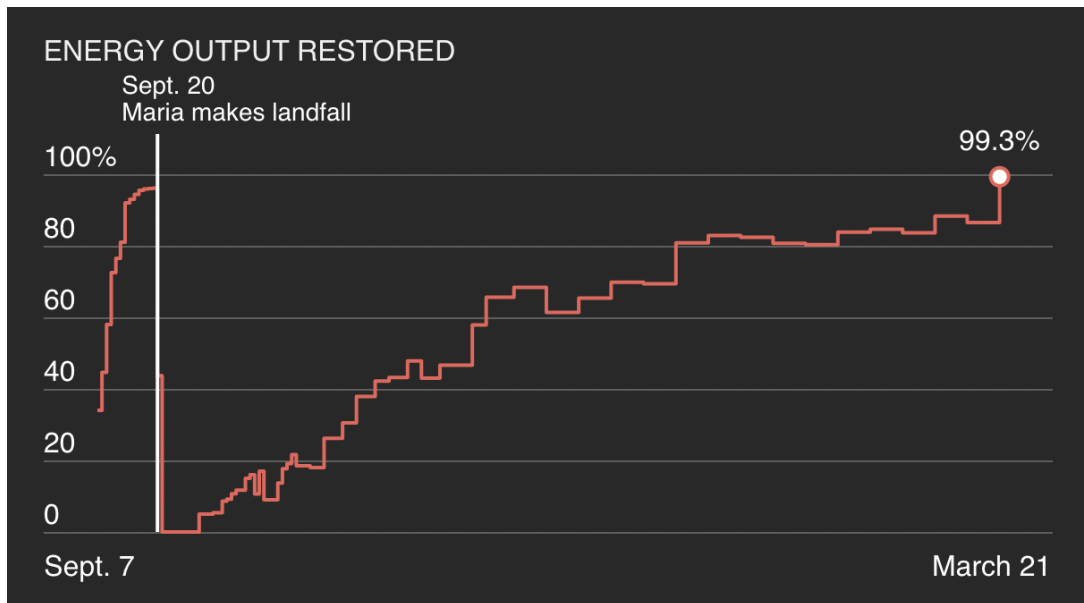


Figure 7: Energy Restoration for Puerto Rico, September 2017-March 2018

Private role	Primary insurer	Private insurers play the role of financial intermediary in the NFIP through what is called the “Write Your Own” (WYO) program, which allows the participating insurers to sell the stand-alone standard flood insurance for which they receive an allowance. Outside of the NFIP there is a small insurance market but it represents only a few percentage points of the entire market. There are ongoing discussions about private insurers starting to sell more flood insurance.
	Reinsurer	
	Administration	
Public role	Primary insurer	The federally-run National Flood Insurance Program (NFIP), established by the National Flood Insurance Act of 1968, provides flood insurance coverage to communities that adopt minimum floodplain management policies. Federally-backed flood insurance coverage is available to any property owner in return for mitigation of flood risks by community ¹⁵ regulation of floodplain development. The NFIP does not buy <i>reinsurance</i> and is authorized to borrow from the U.S. Treasury, as it has done in several occasion (the program is currently \$24 billion in debt).
	Reinsurer	
	Guarantor	
	Regulator	
Public/ private interaction	Coordinated	
	Coexistence	
Voluntary vs. mandatory	Voluntary	The purchase of flood insurance is voluntary however, in 1973, regulations mandated flood insurance purchase for properties in 100-year floodplains with mortgage from a federally backed or regulated lender. ¹⁶
	Mandatory purchase	
	Mandatory offer	
Single vs. bundled	Single	Flood insurance is only available as a stand-alone insurance.
	Bundled	
Premium setting	Risk-based*	The flood insurance premium rates are based on Flood Insurance Rate Maps (FIRMs) produced by FEMA (federal government). However, about one-quarter of the total NFIP insurance contracts (about 1.2 out of 5.3 million policies) are subsidized and their premiums are, on average, 35–50% of the actual risk.
	Partially risk-based	
	Minimally/not risk-based	
	Premium subsidies	
Link between insurance and risk reduction efforts	Warranties/must-do-clauses	Mitigation planning and implementation of risk-reduction and prevention measures are integrated explicitly in the NFIP. Through the Community Rating System (CRS), NFIP’s voluntary program, flood insurance premium rates are discounted to reflect reduced flood risk resulting from creditable community activities. ¹⁷
	Premium/deductible/indemnity limit	
	Awareness-raising/client consulting	
Ex-post policy	Disaster relief fund	For presidentially declared disasters, property owners can receive financial assistance from the federal government through individual assistance grant or low interest loans and are required to purchase flood insurance thereafter.
	Ad-hoc aid	

Figure 8: US Insurance Overview

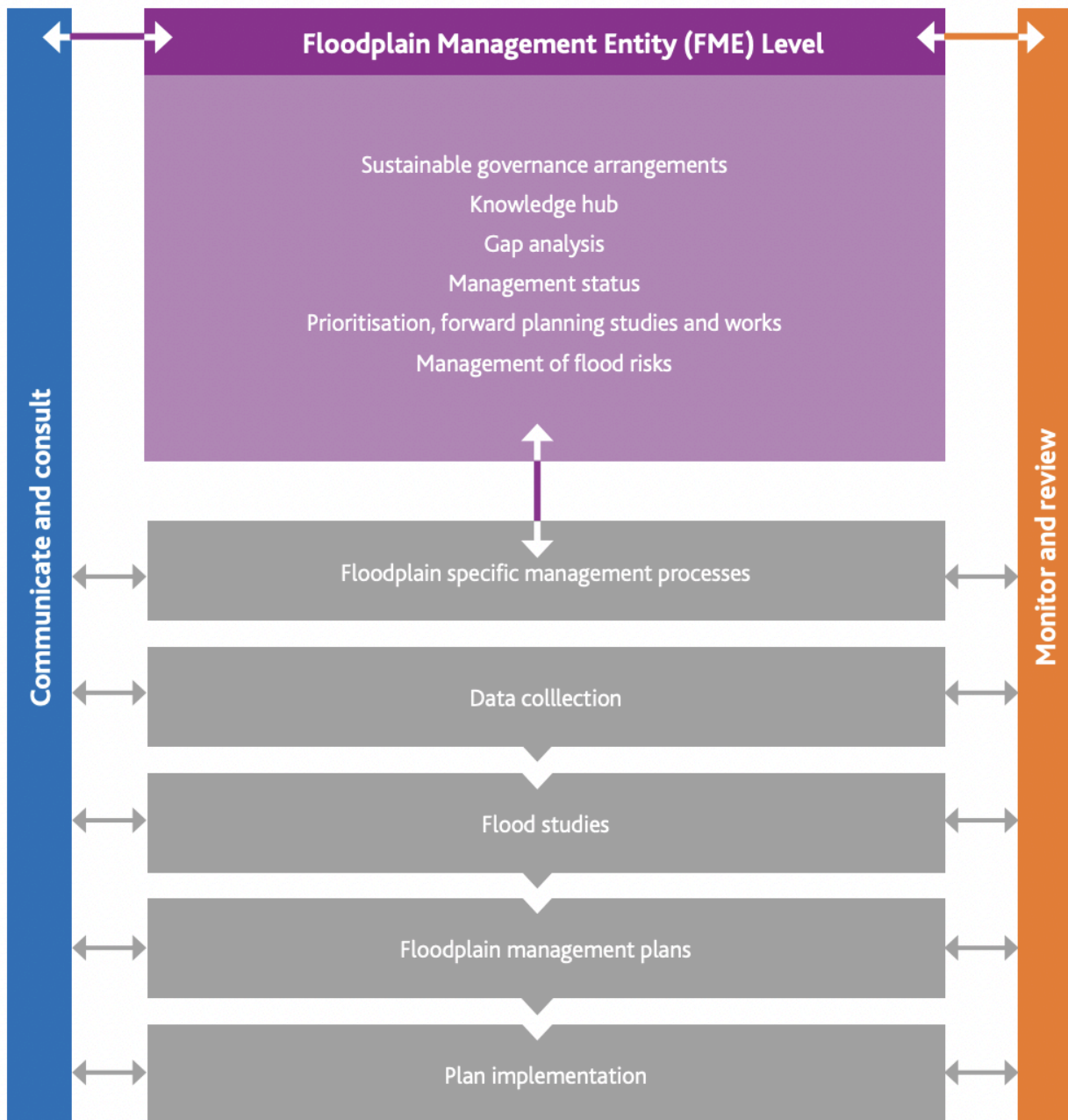
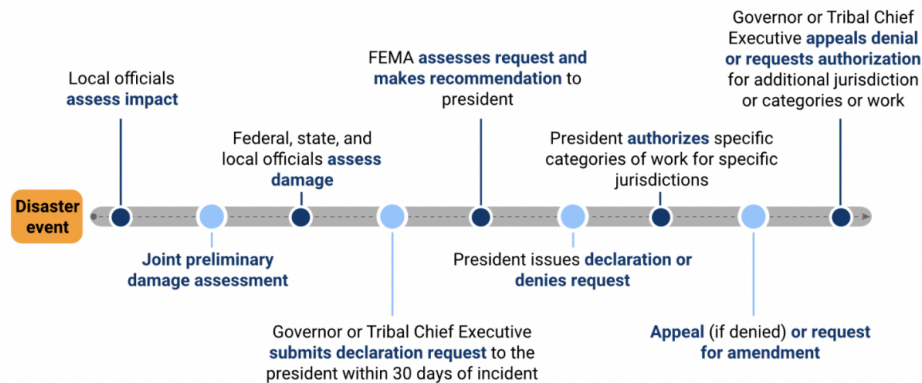


Figure 9: Australia's Flood Risk Management Framework

Disaster declaration request process



Source: Infographic based on timelines developed by Congressional Research Services. The timeline summarizes provisions in 44 C.F.R. §§206.31-206.48

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Figure 10: Disaster Declaration Request Process, US Stafford Act

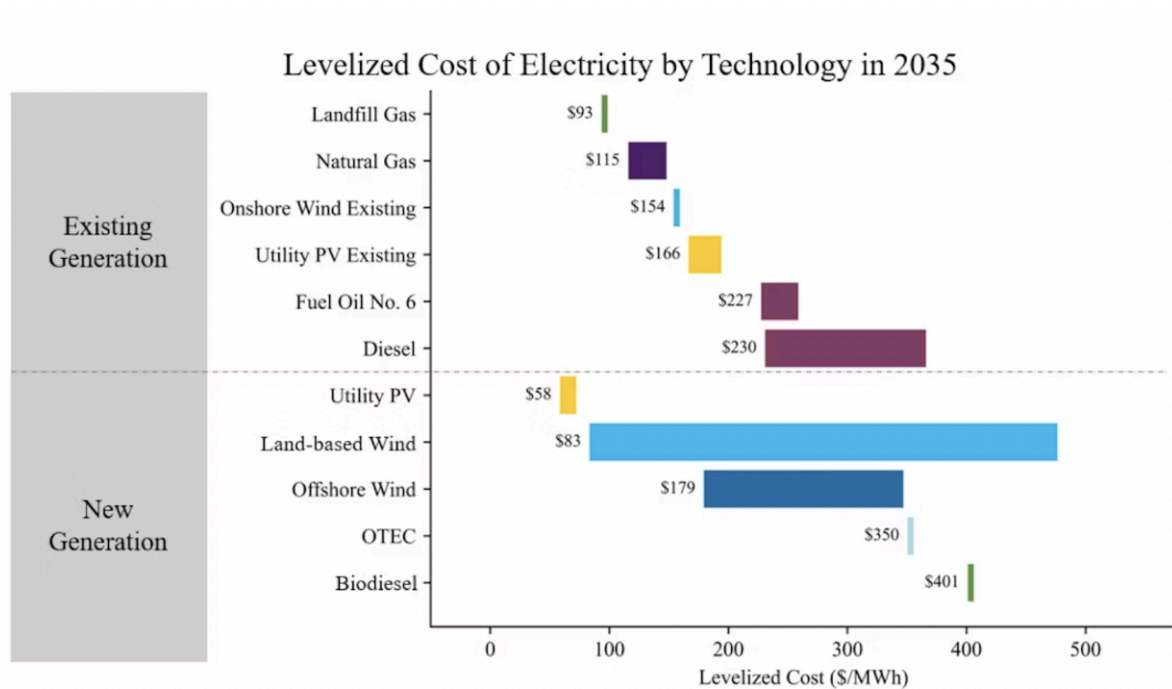


Figure 11: Levelized Cost of Electricity by Technology in 2035, Puerto Rico

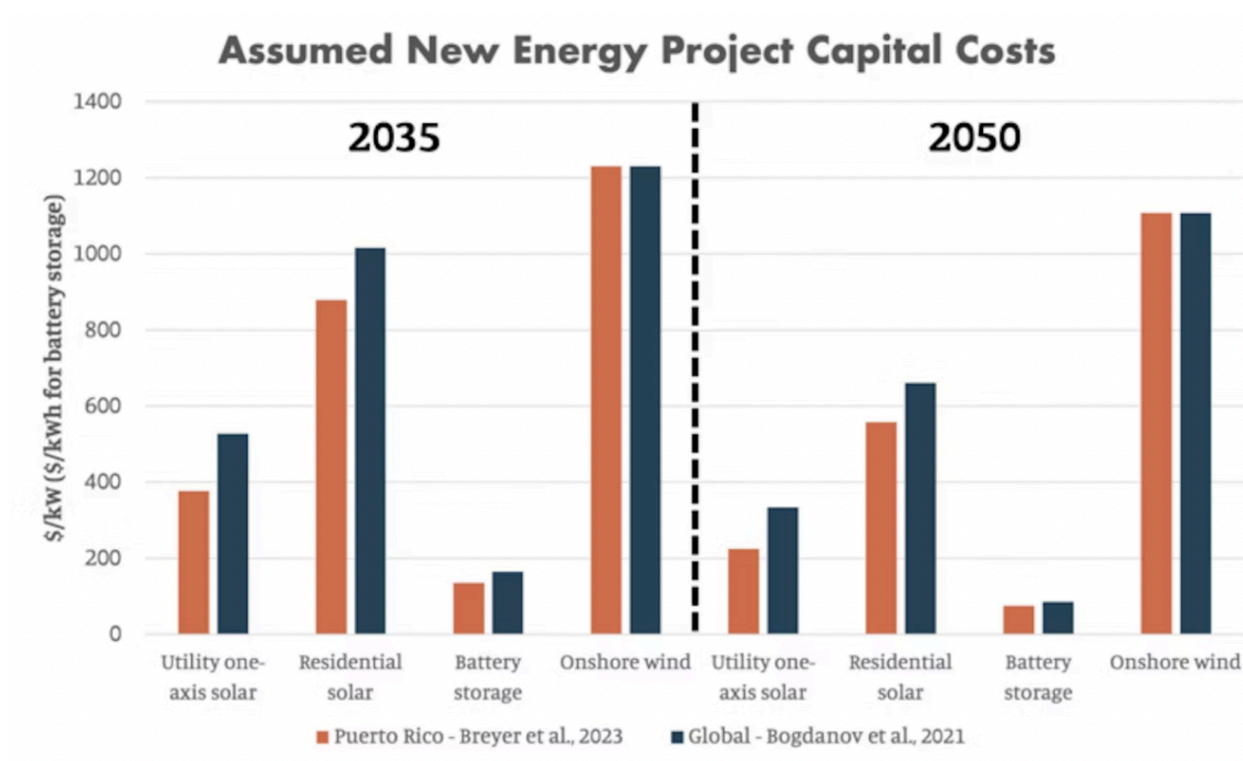


Figure 12: Cost Assumptions - Puerto Rico versus Global

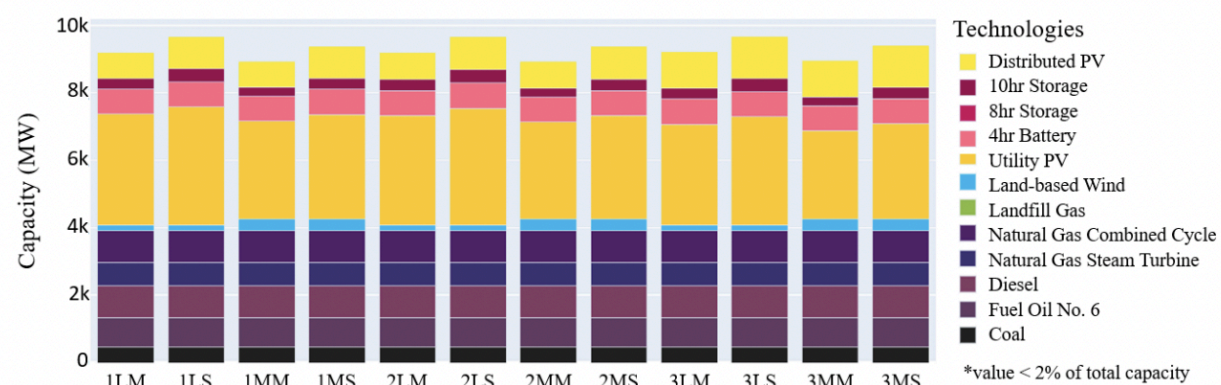


Figure 13: Total Capacity to achieve 40% renewable energy in Puerto Rico

